

LABWORK 6

This lab aims to help you practice pointers, arrays, and functions in C by solving real-life inspired problems. You will implement 10 functions, each focusing on different array manipulations and pointer usage.

Function 1: is_symmetric_temperature (5 pts)

- **Return Type:** `int`
 - **Parameters:** `const int A[], int size`
 - **Description:** Check if an array of daily temperatures is symmetric (morning vs evening). Returns 1 if symmetric, 0 otherwise.
 - **Example:**
 - Input: {20, 22, 24, 26, 26, 24, 22, 20}
 - Output: 1
-

Function 2: rearrange_transactions (5 pts)

- **Return Type:** `void`
 - **Parameters:** `int dest[], const int src[], int size`
 - **Description:** Rearrange daily financial transactions: negatives (debt) on the left, zeros in the middle, positives (income) on the right.
 - **Example:**
 - Input: {500, -200, 0, 100, -50, 0, 300, -100}
 - Output: {-200, -50, -100, 0, 0, 500, 100, 300}
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Function 3: left_shift_steps (5 pts)

- **Return Type:** `void`
 - **Parameters:** `int A[], int size`
 - **Description:** Shift the daily step counts to the left by one. Zero enters from the right.
 - **Example:**
 - Input: {1000, 2000, 1500, 1200}
 - Output: {2000, 1500, 1200, 0}
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Function 4: left_n_shift_schedule (10 pts)

- **Return Type:** `void`
 - **Parameters:** `int A[], int size, int n`
 - **Description:** Shift the daily schedule array n times to the left.
 - **Example:**
 - Input: {1,2,3,4,5}, n=2
 - Output: {3,4,5,0,0}
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Function 5: print_unique_inventory (10 pts)

- **Return Type:** `void`
 - **Parameters:** `const int A[], int size`
 - **Description:** Print unique items in an inventory array (duplicates ignored).
 - **Example:**
 - Input: {5,10,5,3,3,8,10,1}
 - Output: 5 10 3 8 1
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Function 6: compute_student_averages (10 pts)

- **Return Type:** `void`
- **Parameters:** `const int Exams[][4], int students`
- **Description:** Compute and print the average grade for each student using weights: midterms 20%, final 40%.
- **Example:**
 - Input: {{50,60,70,80},{40,50,60,70},{30,40,50,60}}
 - Output:

```
Student 1 avg: 65.00
Student 2 avg: 55.00
Student 3 avg: 45.00
```

Function 7: swap_accounts (10 pts)

- **Return Type:** `void`
 - **Parameters:** `int *a, int *b`
 - **Description:** Swap balances of two bank accounts using pointers.
 - **Example:**
 - Input: a=1000, b=2000
 - Output: a=2000, b=1000
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Function 8: find_first_occurrence (10 pts)

- **Return Type:** `int*`
 - **Parameters:** `int A[], int n, int size`
 - **Description:** Returns a pointer to the first occurrence of n in the array. Returns NULL if not found.
 - **Example:**
 - Input: {5,2,5,3}, n=5
 - Output: Pointer to index 0
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Function 9: find_kth_occurrence (10 pts)

- **Return Type:** `int*`
 - **Parameters:** `int A[], int n, int size, int k`
 - **Description:** Returns a pointer to the kth occurrence of n in the array using find_first_occurrence.
 - **Example:**
 - Input: {5,2,5,3,5}, n=5, k=3
 - Output: Pointer to index 4
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Function 10: count_elements_between_occurrences (10 pts)

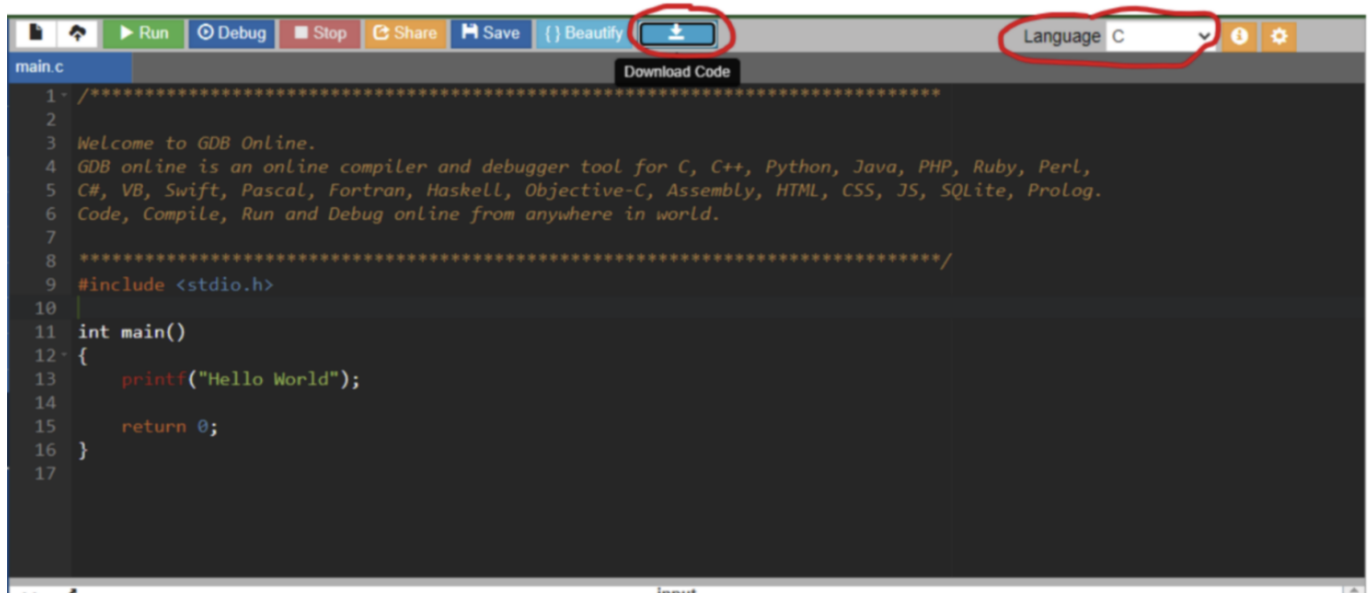
- **Return Type:** `int`
 - **Parameters:** `int A[], int size, int a, int b, int n, int k`
 - **Description:** Count how many times element k appears between the ath and bth occurrence of n in array A.
 - **Example:**
 - Input: A={5,2,5,3,5,2,5}, a=2, b=4, n=5, k=2
 - Output: 1
-

Main Function (15 pts)

- **Function Name:** `main`
 - **Return Type:** `int`
 - **Parameters:** `none`
 - **Description:** Create example arrays for temperatures, transactions, steps, schedules, inventory, exams, accounts, and sales.
 - **Example:**
 - Input: {20, 22, 24, 26, 26, 24, 22, 20}
 - Output: 1
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Submission Format

- Submit your `.c` file named as:
`name_surname.c`



```
1  /******  
2  
3  Welcome to GDB OnLine.  
4  GDB online is an online compiler and debugger tool for C, C++, Python, Java, PHP, Ruby, Perl,  
5  C#, VB, Swift, Pascal, Fortran, Haskell, Objective-C, Assembly, HTML, CSS, JS, SQLite, Prolog.  
6  Code, Compile, Run and Debug online from anywhere in world.  
7  
8  *****/  
9  #include <stdio.h>  
10  
11  int main()  
12  {  
13      printf("Hello World");  
14  
15      return 0;  
16  }  
17
```

Important Notes

- Late submissions will not be accepted.
- Follow function naming and signature exactly.
- Modularize your code using functions.
- Test each function with real-life examples.
- Use loops, pointers, and array arithmetic; do not use extra libraries beyond `stdio.h` and `string.h`.